How GPT-2 Learns to Be a Predictive Coder: Evidence from Directional Activation Patching

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Abstract

To build upon the findings of our first paper—which suggested that transformer-based language models may implement predictive coding (PC) through training—we investigated whether the inter-layer information flow in transformers exhibits directional asymmetry, and whether such asymmetry is consistent with the structure of predictive coding. Using the same six models from our first study and 100 factual semantic pairs, we measured forward patching effects (layer $l \to l{+}k$) and backward patching effects (layer $l{+}k \to l$) to compute a directional asymmetry index. To determine whether any observed asymmetry reflects a learned structure rather than an inherent architectural property, we compared trained GPT-2 Small against an untrained, randomly initialized GPT-2 Small as a control. We find that the GPT-2 family exhibits statistically significant backward asymmetry in the convergence zone ($p < 0.001$ for all three models, $n = 90$), while the randomly initialized model shows a near-zero asymmetry index, indicating that this pattern emerges through training rather than architecture. Furthermore, GPT-Neo and Pythia show non-significant convergence zone asymmetry, and the difference between GPT-2 and these models is statistically significant with confidence intervals excluding zero. These results constitute the first experimental evidence that GPT-2 family models develop a partial predictive coding structure through training.

1 Introduction

Predictive coding (PC) is a neuroscientific theory proposing that the brain processes predictions and error signals in a hierarchical manner [Rao and Ballard, 1999]. In our first paper [Yun, 2026], we found that GPT-2 family models exhibit computational tendencies consistent with predictive coding among widely-used transformer-based language models. However, a key limitation of that work was its inability to determine whether these tendencies reflect a structural property inherent to the transformer architecture, or a learned outcome specific to GPT-2 family models.

To address this limitation, the present paper investigates, through directional activation patching experiments, whether the partial predictive coding structure observed in GPT-2 family models operates as a result of training. We use *partial* predictive coding to indicate that only directional asymmetry consistent with top-down prediction is observed; we do not claim explicit error signal routing or full PC circuit implementation. Key contributions:

1. **Directional asymmetry measurement**: We extend activation patching [Meng et al., 2022] to measure forward and backward information flow across transformer layers, introducing a directional asymmetry index.

2. **Learned vs. architectural**: Comparison against a randomly initialized control model demonstrates that the observed asymmetry is learned through training, not inherent to the transformer architecture.

3. **Training-configuration dependence**: GPT-Neo and Pythia show significantly weaker asymmetry, consistent with the configuration-dependent patterns identified in our first paper.

2 Related Work

2.1 Predictive Coding in Neuroscience and Deep Learning

Rao and Ballard [1999] proposed that the brain minimizes prediction errors via hierarchical top-down predictions and bottom-up error signals. Millidge et al. [2022] proved that PC is mathematically equivalent to backpropagation, implying that backpropagation-trained networks may implicitly implement PC-like computations.

2.2 Mechanistic Interpretability

Elhage et al. [2021] introduced the residual stream framework for principled analysis of information flow in transformers. Meng et al. [2022] showed via activation patching that factual associations are localized to specific MLP layers—a methodology we extend here to measure directional asymmetry.

2.3 PC-Consistent Tendencies in GPT-2

Yun [2026] provided multi-method empirical evidence that GPT-2 family models exhibit PC-consistent computational tendencies, while leaving open the question of whether these tendencies are architectural or learned. The present work directly addresses this question.

3 Methods

3.1 Models

We use the same six models as in our first paper: GPT-2 Small (117M), GPT-2 Medium (345M), and GPT-2 Large (774M) as primary subjects, with GPT-Neo 125M and Pythia 160M/410M as architectural controls. All models were loaded via TransformerLens [Nanda and Bloom, 2022] with frozen weights on a Google Colab T4 GPU. Code is available at `https://github.com/gallam-research-dev/gallam-research-dev-gpt2-predictive-coder`. To isolate the effect of training from architecture, we additionally include a randomly initialized GPT-2 Small as a control, in which all weights are reinitialized from a normal distribution ($\mu = 0.0$, $\sigma = 0.02$) matching GPT-2's original initialization scheme.

Table 1: Model configurations used in this study.

Model	Params	Layers	Training Data	Tokenizer
GPT-2 Small	117M	12	WebText	GPT-2 BPE
GPT-2 Medium	345M	24	WebText	GPT-2 BPE
GPT-2 Large	774M	36	WebText	GPT-2 BPE
GPT-Neo 125M	125M	12	The Pile [Gao et al., 2021]	GPT-2 BPE
Pythia 160M	160M	12	The Pile	NeoX
Pythia 410M	410M	24	The Pile	NeoX
GPT-2 Small (Random)	117M	12	None (control)	GPT-2 BPE

3.2 Semantic Pairs

We construct 100 factual semantic pairs across seven categories: capital cities (20), currencies (10), languages (15), colors (12), antonyms (15), scientific facts (15), and occupations/locations (13). Each pair consists of a clean prompt with a correct target token and a corrupt prompt with a foil token. Only pairs where both target and foil are single tokens under the model's tokenizer are retained, yielding n=90 valid pairs for GPT-2 variants and 84–90 for control models.

3.3 Directional Activation Patching

For each valid semantic pair, we run a clean forward pass and cache all residual stream activations. We then define layer pairs (l, $l{+}k$) where k=3, and for each pair conduct two patching experiments:

Forward patching: The clean activation at layer l is injected into the corrupt run at layer $l{+}k$.

Backward patching: The clean activation at layer $l{+}k$ is injected into the corrupt run at layer l.

The patching effect for each direction is:

$$\text{Effect} = \text{logit}_{\text{patched}}(\text{target}) - \text{logit}_{\text{patched}}(\text{foil}) - \text{baseline}, \tag{1}$$

where baseline is the logit difference of the unpatched corrupt run [Meng et al., 2022]. The directional asymmetry index is:

$$\text{Asymmetry} = \text{Effect}_{\text{backward}} - \text{Effect}_{\text{forward}}. \tag{2}$$

A positive asymmetry index indicates that backward (top-down) patching has stronger causal influence than forward (bottom-up) patching.

3.4 Zone Definitions

Following Yun [2026], we define three functional zones:

Table 2: MLP zone definitions.

Zone	Normalized Position	Hypothesized PC Role
Early Spike	0.10–0.20	Initial error signal processing
Convergence	0.25–0.75	Error minimization
Late Spike	0.80–0.95	Final prediction output

3.5 Statistical Analysis

For each model, we compute a pair-level asymmetry score by averaging the asymmetry index across all layer pairs within the convergence zone for each semantic pair. This yields n=90 observations for GPT-2 variants and n=84–90 for control models. We apply one-sample t-tests against zero and report 95% bootstrap confidence intervals (n=10,000 resamples, seed = 42). For between-group comparisons (trained vs. random-initialized; GPT-2 family vs. GPT-Neo/Pythia), we report bootstrap confidence intervals of the mean difference, treating non-overlap with zero as the significance criterion.

4 Results

4.1 Directional Asymmetry in GPT-2 Family

All three GPT-2 models exhibit statistically significant positive asymmetry in the convergence zone (normalized position 0.25–0.75), indicating that backward patching produces stronger causal effects than forward patching in this region.

Table 3: Pair-level convergence zone asymmetry (one-sample t-test vs. 0). n = number of valid semantic pairs per model. Sig = *** if $p < 0.001$.

Model	Mean	95% CI	t	p	Sig	n
GPT-2 Small (117M)	+0.969	[+0.697, +1.252]	+6.850	< 0.001	***	90
GPT-2 Med (345M)	+0.290	[+0.167, +0.421]	+4.481	< 0.001	***	90
GPT-2 Large (774M)	+0.194	[+0.090, +0.300]	+3.565	< 0.001	***	90

Within the convergence zone, asymmetry is initially negative or near zero in the early portion (0.25–0.50), then transitions to strongly positive in the later portion (0.55–0.75), consistent with a directional shift from forward-dominant to backward-dominant processing.

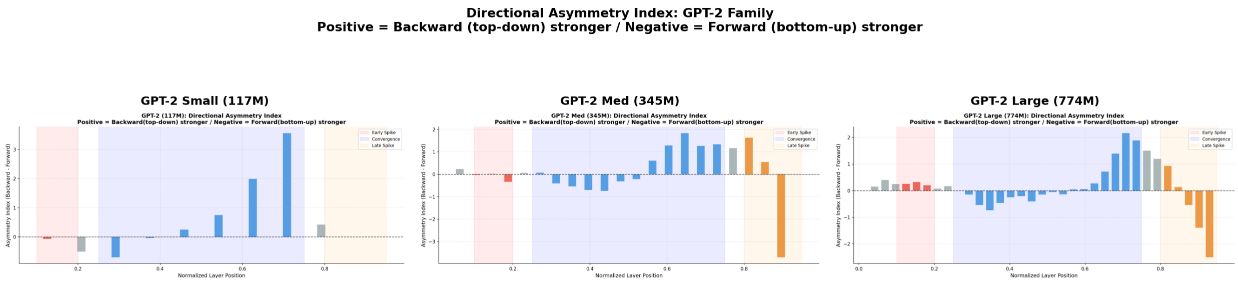

Figure 1: Directional asymmetry index by normalized layer position across all six models. Positive values indicate backward (top-down) patching dominance; negative values indicate forward (bottom-up) dominance. Shaded regions: Early Spike (red), Convergence (blue), Late Spike (orange).

4.2 Random-Initialized Control

The randomly initialized GPT-2 Small shows a near-zero asymmetry index across all zones (convergence zone mean = −0.008, 95% CI [−0.080, +0.063], n.s.), indicating no systematic directional preference. Comparison between trained and random-initialized models reveals statistically significant differences for all three GPT-2 variants.

Table 4: Trained vs. random-initialized GPT-2 comparison (convergence zone). * = bootstrap 95% CI excludes zero. n_{trained}=90, n_{random}=90.

Model	Diff	95% CI	Sig	Cohen's d
GPT-2 Small (117M)	+0.977	[+0.696, +1.265]	*	1.00
GPT-2 Med (345M)	+0.298	[+0.156, +0.446]	*	0.60
GPT-2 Large (774M)	+0.202	[+0.075, +0.332]	*	0.46

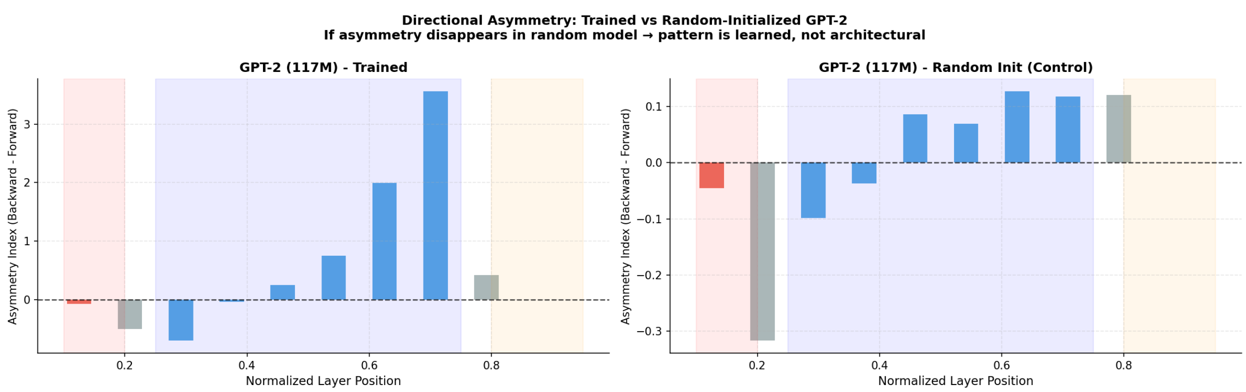

Figure 2: Directional asymmetry index: trained GPT-2 Small (left) vs. randomly initialized GPT-2 Small (right). The near-zero asymmetry in the random model confirms that the pattern is learned, not architectural.

4.3 GPT-Neo and Pythia

GPT-Neo 125M and Pythia 160M/410M show non-significant convergence zone asymmetry (GPT-Neo: mean $= -0.345$, n.s.; Pythia 160M: mean $= -0.105$, n.s.; Pythia 410M: mean $= -0.128$, n.s.), with asymmetry indices trending negative rather than positive. Comparison between the GPT-2 family and each control model reveals statistically significant differences with confidence intervals excluding zero in all three cases.

Table 5: GPT-2 family vs. GPT-Neo/Pythia comparison (convergence zone). * = bootstrap 95% CI excludes zero.

Comparison	Diff	95% CI	Sig	Cohen's d
GPT-2 family vs GPT-Neo (125M)	+0.830	[+0.413, +1.260]	*	0.53
GPT-2 family vs Pythia (160M)	+0.589	[+0.333, +0.848]	*	0.57
GPT-2 family vs Pythia (410M)	+0.612	[+0.426, +0.804]	*	0.73

5 Discussion

The experimental results show that trained GPT-2 family models exhibit significantly stronger backward asymmetry than forward asymmetry in the convergence zone, and a comparison between trained GPT-2 Small and randomly initialized GPT-2 Small yields a Cohen's d of 1.00, indicating a large effect size. Furthermore, given that the same tendency is not observed in GPT-Neo and Pythia—which share the same transformer architecture—we conclude that partial predictive coding emerges as a result of training in GPT-2 family models.

Intra-zone asymmetry pattern. The convergence zone shows a two-phase pattern: forward-dominant in the early portion (0.25–0.50) and backward-dominant in the later portion (0.55–0.75). This intra-zone reversal is itself noteworthy and warrants further investigation. A simple PC interpretation would predict error-minimization throughout the convergence zone, but the forward-dominant early phase suggests a more complex processing structure that the present data cannot fully resolve.

Effect size decreases with model scale. Cohen's d for the trained-vs-random comparison decreases monotonically with model size: Small (1.00) $>$ Medium (0.60) $>$ Large (0.46). This inverse scaling is an interesting pattern that runs counter to the expectation that larger models would show stronger PC-consistent structure. One possible explanation is that larger models distribute the relevant computation across more layers, diluting the per-pair signal. This observation deserves dedicated follow-up.

GPT-Neo shows directional reversal, not merely non-significance. GPT-Neo's convergence zone mean of -0.345 is not only non-significant but directionally reversed relative to GPT-2, trending toward forward-dominance. This is a qualitatively different pattern from non-significance alone, and may indicate that GPT-Neo implements a fundamentally different information-flow structure. Future work should distinguish between the absence of PC structure and the presence of an alternative directional organization.

Interpretive caution on backward patching. We note that backward patching (injecting layer $l+k$ activations into layer l) does not directly measure top-down information flow in the forward pass. Rather, it measures the causal influence of higher-layer representations when artificially introduced at lower layers. The interpretation of this as evidence for a top-down predictive signal is suggestive but not conclusive: the same pattern could arise from residual stream accumulation in a purely feedforward model. This limitation is acknowledged and motivates future experiments that directly isolate residual stream contributions.

It remains unclear whether the divergence observed in GPT-Neo and Pythia stems from differences in training data (WebText vs. The Pile [Gao et al., 2021]), architectural details, or both. Finally, the present findings are limited to the GPT-2 family and may not generalize to other model families.

6 Conclusion

In conclusion, we present experimental evidence that partial predictive coding tendencies emerge as a result of training differences in GPT-2 family models. If these tendencies are attributable to training data composition, investigating whether such tendencies can be artificially induced in other models constitutes a compelling direction for future research.

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A Supplementary Figures

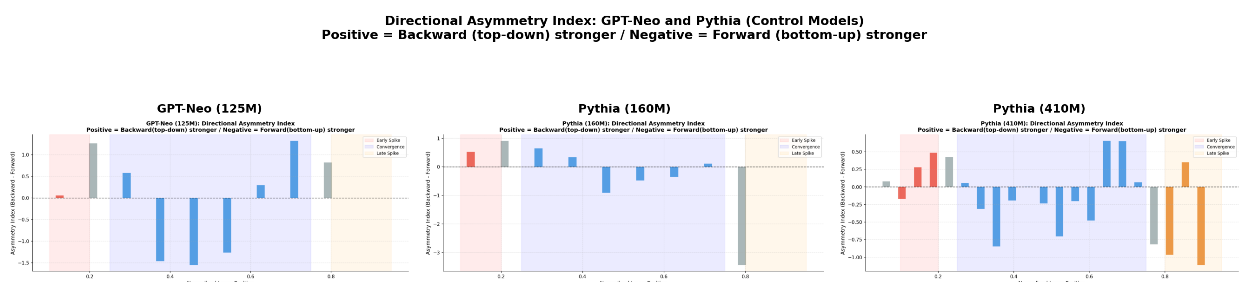

Figure 3: Directional asymmetry index for GPT-Neo (125M), Pythia (160M), and Pythia (410M). All three models show non-significant or negative convergence zone asymmetry, in contrast to the GPT-2 family.